

RATE & EXTENT – CATALYSTS

KEY VOCABULARY

Catalyst: speeds up a reaction without being used up.

Activation energy: the minimum energy for a successful collision.

Alternative pathway: a different route with a lower activation energy.

Enzyme: a biological catalyst.

COMMON MISCONCEPTIONS

~~— A catalyst is used up in the reaction.~~

✓ A catalyst is unchanged and can be reused.

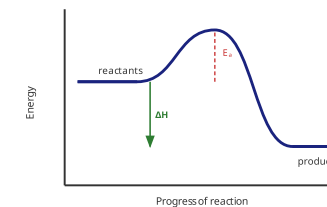
~~— A catalyst raises activation energy.~~

✓ A catalyst LOWERS the activation energy.

~~— A catalyst increases the yield.~~

✓ It changes the rate, not the amount of product.

PROFILE WITH CATALYST



A catalyst gives a route with a **lower E_a** (smaller hump).

1 DEFINE CATALYST

State what is meant by a **catalyst**. [1 mark]

2 HOW IT WORKS

Explain how a catalyst increases the rate of a reaction. [2 marks]

3 NOT USED UP

Explain why a catalyst is not included in the chemical equation. [2 marks]

4 ENZYMES

Name the type of catalyst found in living things. [1 mark]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

A catalyst provides an alternative ____ with a lower activation energy, which ____ the energy needed.

lowers

pathway

raises

mass

6 ECONOMIC BENEFIT

Suggest why catalysts are valuable in industry. [2 marks]